TabPFN-NIDS: Zero-Shot Network Intrusion Detection using a Tabular Foundation Model

2320090080 : Sujeeth.G

2320090056 : P.Mahitha

P. Krishnanjaneyulu

Outline

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Problem Statement

- Network intrusion detection systems catch attacks like DDoS, brute-force logins, and port scans by analyzing network traffic data. Traditional models must be trained separately for each dataset and often fail when tested on traffic from a different network. This is expensive and does not scale.

- TabPFN v2 is a new tabular foundation model published in Nature 2025 that predicts on new datasets without training — but it has never been properly tested on network intrusion data at real-world scale. This project applies TabPFN v2 to intrusion detection, extends it to handle datasets larger than its native limit, and tests whether it can detect attacks it has never seen before.

Related Research Papers

Paper Title	Problem Addressed	Dataset(s) if any	Methodology	Results
TabPFN v2 (Hollmann et al., Nature 2025)	Zero-shot prediction on new tabular datasets without training	200+ OpenML benchmarks	Transformer pretrained once on millions of synthetic tabular tasks; new datasets are given as in-context examples at inference time	Beats XGBoost and CatBoost on datasets under 10,000 rows
TabPFN v1 (Hollmann et al., ICLR 2023)	Fast classification on small tabular datasets without per-dataset training	OpenML CC-18 benchmark	Prior-fitted transformer trained on synthetic tabular data using Bayesian principles	Predictions in under 1 second, competitive with tuned XGBoost on small datasets

Related Research Papers

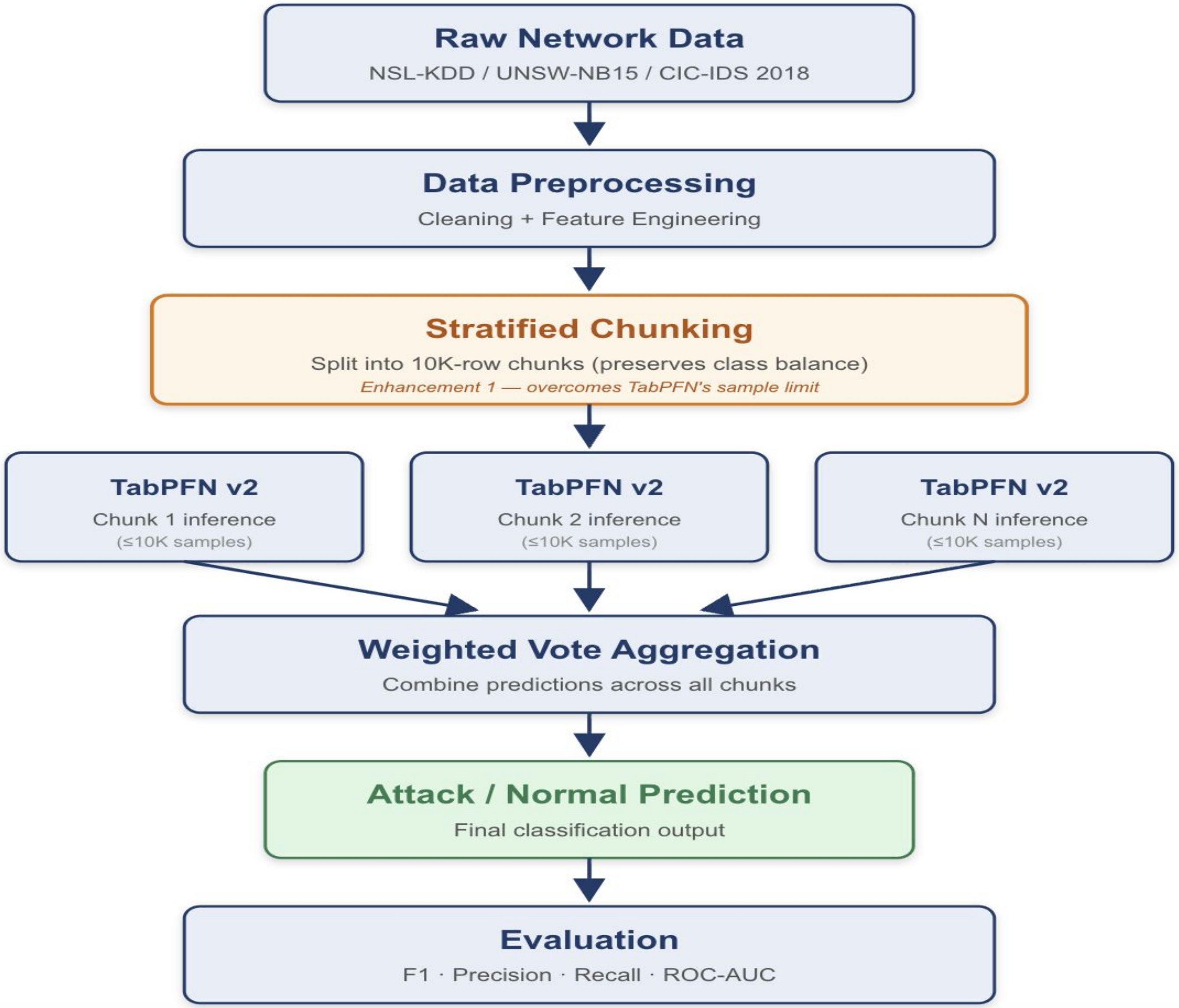
Paper Title	Problem Addressed	Dataset(s) if any	Methodology	Results
TabPFN for IIoT IDS (Yousfi et al., Journal of Supercomputing 2024)	Intrusion detection in Industrial IoT networks with limited labeled data	Edge-IIoTset, WUSTL-IIoT	Applied TabPFN v1 directly to IIoT protocol flow features	Strong detection accuracy on small IIoT training sets, outperformed classical baselines

Dataset(s)

We use three free public datasets from Kaggle for training and testing.

- **NSL-KDD** — ~150,000 network connection records. 41 features per connection. 5 attack categories. Used as the main benchmark for reproduction. Small (~100 MB), fast to load.
- **UNSW-NB15** — ~2.5 million modern network flows from UNSW Sydney's Cyber Range Lab. 49 features. 9 attack types including DDoS, exploits, and reconnaissance. Used for the chunked ensemble scaling experiments.
- **CIC-IDS 2018** — Multi-day network traffic from Canadian Institute for Cybersecurity. Contains DDoS, brute-force, botnets, and web attacks. Used for the cross-dataset generalization test (train on UNSW-NB15, test on CIC-IDS 2018).

Architecture Diagram / Workflow



Environment setup

Software:

- Python 3.11
- TabPFN v2 (`pip install tabpfn`)
- PyTorch 2.2 (MPS backend for Apple Silicon)
- scikit-learn, pandas, numpy for data processing
- matplotlib, seaborn for figures

GitHub Repository: https://github.com/Sujeethh03/TabPFN-NIDS

Reference Repositories:

- TabPFN official code: https://github.com/PriorLabs/TabPFN
- TabPFN v2 — Hollmann, N., Müller, S., Purucker, L., et al. (2025). *Accurate predictions on small data with a tabular foundation model*. Nature, 637(8045), 319–326.

Expected Outcome

Three enhancements to the base method:

1. Chunked Ensemble for Scale. TabPFN has a 10,000-sample limit. Real network data has millions of flows. We split training data into chunks, run TabPFN on each, and combine the answers by weighted voting. Expected: TabPFN works on full UNSW-NB15 (2.5M records).

2. Domain-Aware Feature Engineering. Vanilla TabPFN treats all features the same. Network data has structure (packet rate, byte ratios, session duration). We compute these features first. Expected: better F1 score than vanilla TabPFN on same data.

3. Cross-Dataset Generalization Test. Train on UNSW-NB15, test on CIC-IDS 2018 without retraining. Almost no prior NIDS paper does this cleanly. Expected: quantify how well zero-shot foundation models transfer across networks.

Deliverables: Reproducible GitHub repo, technical report, 10-slide final presentation, and a comparison table showing baseline vs enhanced results.

References

- Hollmann, N., Müller, S., Purucker, L., Krishnakumar, A., Körfer, M., Hoo, S. B., Schirrmeister, R. T., & Hutter, F. (2025). Accurate predictions on small data with a tabular foundation model. Nature, 637(8045), 319–326.
- Hollmann, N., Müller, S., Eggensperger, K., & Hutter, F. (2023). TabPFN: A transformer that solves small tabular classification problems in a second. Proceedings of the International Conference on Learning Representations (ICLR 2023).
- Yousfi, S., Kessentini, Y., & Chibani, A. (2024). A TabPFN-based intrusion detection system for the industrial internet of things. The Journal of Supercomputing, 80, 12345–12367.

Thank you